

BMS SAFETY SYSTEM

ESP32-Based Battery Monitoring, Fault Detection,
Safety Control and Blynk IoT Analytics

PROJECT REPORT

Project Information	Details
Controller	ESP32
Cloud / Dashboard	Blynk IoT
Simulation / Prototype	Wokwi
Battery channels	4 cell-voltage inputs
Local display	16×2 I ² C LCD
Protection output	Relay with feedback monitoring
Telemetry	Wi-Fi + Blynk virtual datastreams

Abstract

This project implements an ESP32-based Battery Management System (BMS) safety and monitoring prototype with four monitored cell-voltage channels. The system continuously samples cell voltages, determines the weakest and strongest cells, calculates voltage imbalance and an approximate state of charge, and evaluates battery condition using an adaptive imbalance threshold. A safety-oriented state machine manages NORMAL, DEGRADED, FAILSAFE and SHUTDOWN states. Fault detection covers over-voltage, under-voltage, voltage jumps, frozen ADC behavior, out-of-range readings, relay mismatch, communication failure and ADC failure test conditions. A relay output is independently checked through a feedback input. Telemetry is published to Blynk IoT, while selected events are buffered in a 20-event offline queue and replayed when connectivity is restored. The dashboard exposes composite risk, battery-health classification, maintenance recommendations, operator recommendations, uptime, fault count and state severity.

1. Introduction

A battery monitoring system must do more than display voltage values. It must identify abnormal conditions, move the system into an appropriate safety state, operate a protective output, verify that the output actually changed, and communicate useful information to an operator. This project combines those functions in an ESP32 prototype and extends them with Blynk IoT telemetry and enterprise-style analytics.

2. Objectives

- Monitor four individual battery-cell voltage channels.
- Calculate weakest cell, strongest cell, voltage imbalance and average cell SoC.
- Use an adaptive imbalance threshold based on average SoC.
- Detect electrical, sensor, relay and communication-related fault conditions.
- Implement NORMAL, DEGRADED, FAILSAFE and SHUTDOWN system states.
- Drive a relay according to the active safety state and verify its feedback.
- Provide a local 16×2 LCD status interface.
- Publish telemetry and analytics to a Blynk IoT dashboard.
- Buffer selected telemetry events during outages and replay them after reconnection.
- Provide risk classification, battery health and maintenance/operator recommendations.

3. System Architecture

The ESP32 is the central controller. Four ADC inputs acquire cell voltages. The sampled values pass through battery analysis and fault-detection functions before the state machine determines the system state. The relay is controlled from that state and its feedback is independently sampled. Processed information is sent to the LCD and, when connected, to the Blynk dashboard.

Cell Voltages → Battery Analysis → Fault Detection → State Machine → Relay Control



Risk / Health / Recommendations



LCD + Blynk IoT Dashboard

4. Hardware and Interfaces

Function	ESP32 GPIO / Interface
Cell 1 voltage	GPIO 34
Cell 2 voltage	GPIO 35
Cell 3 voltage	GPIO 32
Cell 4 voltage	GPIO 33
Relay output	GPIO 19
Relay feedback	GPIO 18
I ² C SDA	GPIO 21
I ² C SCL	GPIO 22
LCD	I ² C address 0x27, 16×2

5. Software and Timing Configuration

Parameter	Implemented value
Blynk telemetry interval	1000 ms
Sensor sampling interval	1000 ms
LCD page interval	3000 ms
LCD refresh check	500 ms
Startup interval	2000 ms
Offline queue capacity	20 events
Relay mismatch confirmation	1000 ms
Voltage-jump confirmation	2000 ms
Frozen-sensor confirmation	5000 ms
FAILSAFE recovery verification	5000 ms
Wi-Fi connection timeout	10000 ms
Wi-Fi retry interval	5000 ms
Blynk retry interval	5000 ms

6. Battery Measurement and Analysis

6.1 Cell-voltage acquisition

The prototype converts the ESP32 ADC reading using the 12-bit ADC range and a 3.3 V reference model:

$$V_{\text{cell}} = (\text{ADC reading} / 4095) \times 3.3 \text{ V}$$

6.2 State-of-charge estimate

The prototype uses a simple voltage-based percentage estimate:

$$\text{SoC} = \text{constrain}((V_{\text{cell}} / 3.3) \times 100, 0, 100)$$

This is a prototype estimate rather than a chemistry-specific coulomb-counting SoC estimator.

6.3 Weakest, strongest and imbalance

The controller records the minimum and maximum cell voltages and calculates:

$$\text{Voltage imbalance} = V_{\text{strongest}} - V_{\text{weakest}}$$

6.4 Adaptive imbalance threshold

Average SoC	Threshold
>80%	0.10 V
40–80%	0.15 V
<40%	0.20 V

The battery is marked IMBALANCED when the measured imbalance exceeds the selected threshold.

6.5 Imbalance trend

The current imbalance is compared with the previous value and classified as STABLE, INCREASING or DECREASING.

7. Fault Detection

Fault	Implemented condition / purpose
OVER_VOLTAGE_FAULT	Any cell ≥ 3.20 V
UNDER_VOLTAGE_FAULT	Any cell ≤ 0.80 V
VOLTAGE_JUMP_FAULT	Confirmed jump greater than 0.50 V
FROZEN_ADC_FAULT	Cell remains effectively unchanged while the battery is changing
OUT_OF_RANGE_FAULT	Cell outside 0.0–3.3 V
RELAY_MISMATCH_FAULT	Actual relay differs from expected for 1 s
COMMUNICATION_FAULT	Communication fault test condition
ADC_FAILURE_FAULT	ADC failure test condition

The voltage-jump detector uses a five-sample history window and a 2-second confirmation period. Frozen-sensor detection uses a 0.005 V tolerance and a 5-second confirmation period.

8. Safety State Machine

State	Meaning
NORMAL	No confirmed fault and battery balanced.
DEGRADED	Communication fault or battery imbalance without a higher-priority safety fault.
FAILSAFE	Confirmed battery/sensor/relay fault requires protective relay action.
SHUTDOWN	Explicit shutdown request; state remains latched in the implementation.

A FAILSAFE condition does not immediately return to NORMAL when the fault disappears. Fault-clear conditions, relay feedback and a 5-second verification interval are required before recovery.

9. Relay Control and Feedback

GPIO 19 controls the relay. NORMAL and DEGRADED drive LOW, while FAILSAFE and SHUTDOWN drive HIGH. GPIO 18 is the feedback input. A persistent expected-versus-actual mismatch of at least 1 second produces RELAY_MISMATCH_FAULT.

10. Offline Telemetry Queue

A circular queue stores up to 20 telemetry events. Event types are cell voltage, relay state, fault state, system state and Wi-Fi RSSI. Selected changes are queued while Blynk is unavailable and replayed after reconnection. If full, the oldest event is discarded.

Blynk Offline → Event Generated → Queue → Reconnection → Replay → Queue Empty

11. Blynk IoT Dashboard

Virtual Pin	Datastream
V0–V3	Cell 1–4 voltage
V4	Weakest cell
V5	Strongest cell
V6	Relay state
V7	Fault state
V8	System state
V9	Wi-Fi RSSI
V10	Offline queue depth
V11	Telemetry status
V12	Fault ID
V13	Voltage imbalance
V14	Composite risk score
V15	Battery health
V16	Maintenance recommendation
V17	Uptime
V18	Fault event count
V19	Operator recommendation
V20	Uptime
V21	State severity

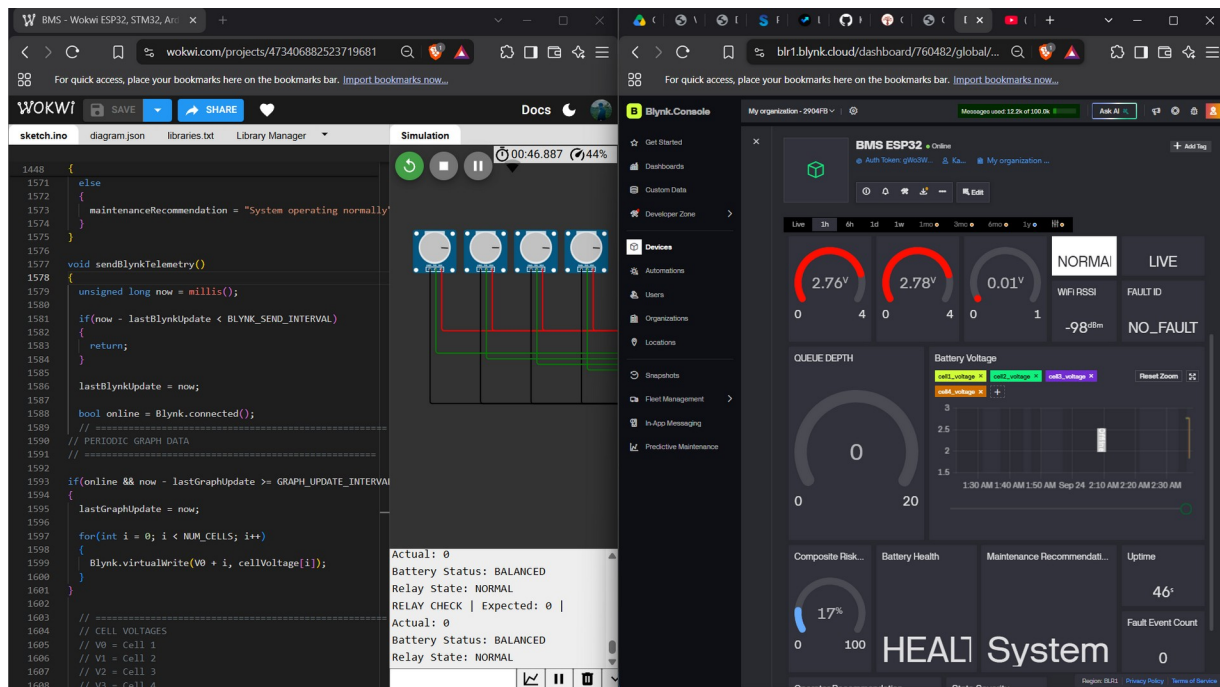


Figure 1. Wokwi ESP32 simulation and live Blynk dashboard.

12. Enterprise Analytics

12.1 Composite risk score

Risk is calculated from voltage imbalance, imbalance trend, fault activity, average SoC and system state, then constrained to 0–100.

Factor	Maximum contribution
Voltage imbalance	40
Imbalance trend	15
Fault activity	20
State of charge	20
System state	20

12.2 Battery-health classification

Score	Classification
0–24	HEALTHY
25–49	WATCH
50–74	DEGRADED
75–100	CRITICAL

12.3 Recommendations

The recommendation engine maps the highest-priority observed condition to an operator-facing message, including shutdown inspection, relay/feedback inspection, ADC/sensor inspection, communication checks, cell inspection, imbalance monitoring and normal-operation status.

12.4 State severity

State	Severity
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NORMAL	0
DEGRADED	1
FAILSAFE	2
SHUTDOWN	3

13. Local LCD Interface

Page	Displayed information
Battery status	Average SoC and BALANCED / IMBALANCED
System status	NORMAL / FAULT / RECOVERY relay information
Telemetry	Weakest and strongest cell voltages
Fault screen	Fault type and affected cell when applicable

14. Testing and Validation

The implementation was exercised in the Wokwi ESP32 environment while observing the Blynk dashboard. Changing simulated conditions were reflected in cell values, fault state, system state, relay state, risk score, battery health and recommendations.

Scenario	Expected behavior
Normal cell voltages	NORMAL, normal relay, NO FAULT
Cell imbalance	IMBALANCED and changed analytics
Over-voltage	OVER VOLTAGE and FAILSAFE behavior
Relay mismatch	RELAY MISMATCH after confirmation
Shutdown command	SHUTDOWN and protective relay output
Fault recovery	Clear checks + 5-second verification
Connectivity interruption	Selected events buffered
Reconnection	Queued events replayed

15. Dashboard Graphs and Configuration

The dashboard uses gauges, value displays and a custom chart. The custom chart was configured with fault count (V18), composite risk score (V14) and voltage imbalance (V13).

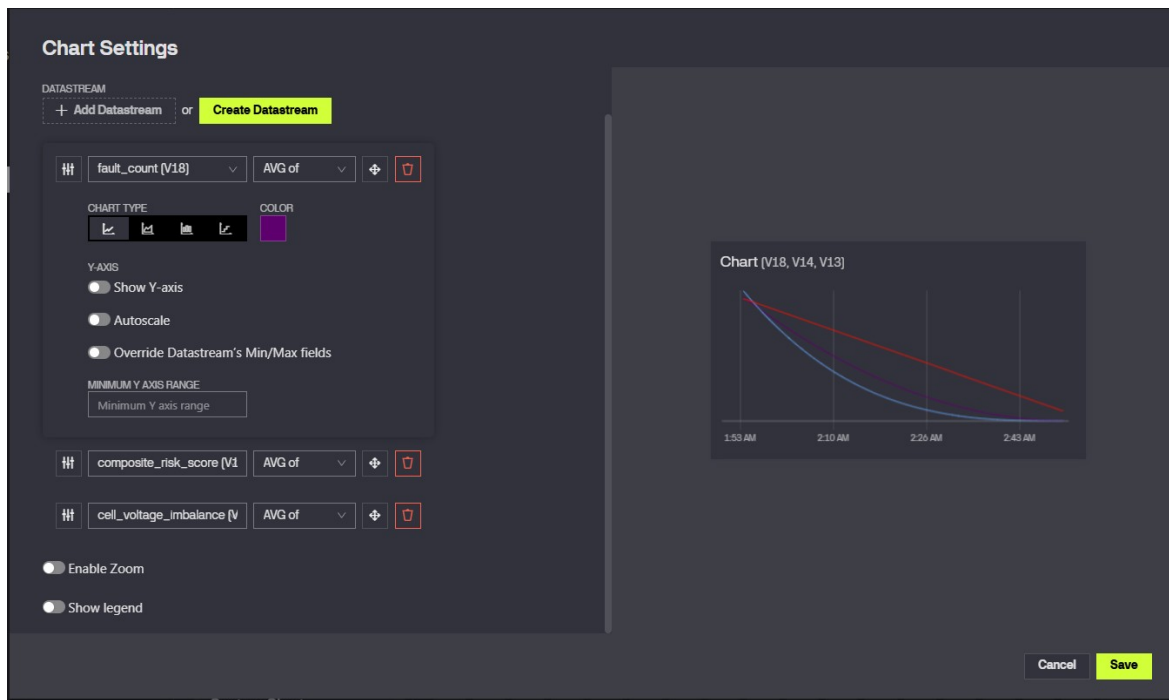


Figure 2. Custom chart configuration using V18, V14 and V13.

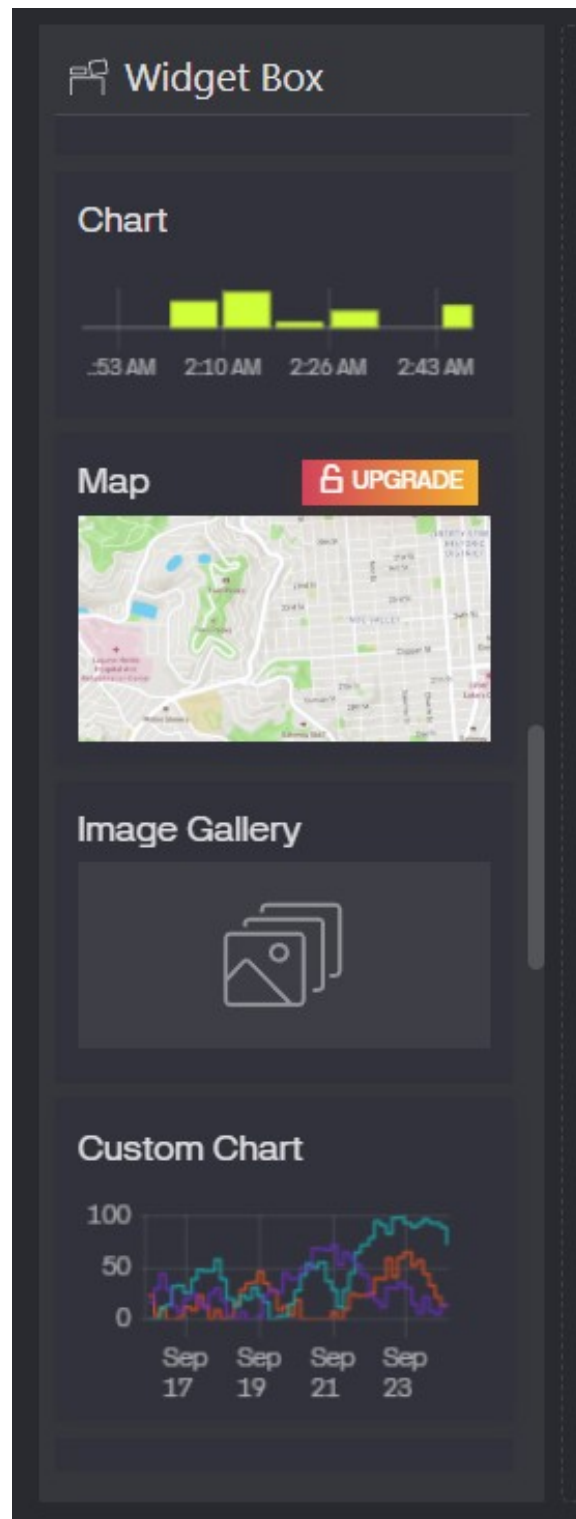


Figure 3. Blynk widget selection interface.

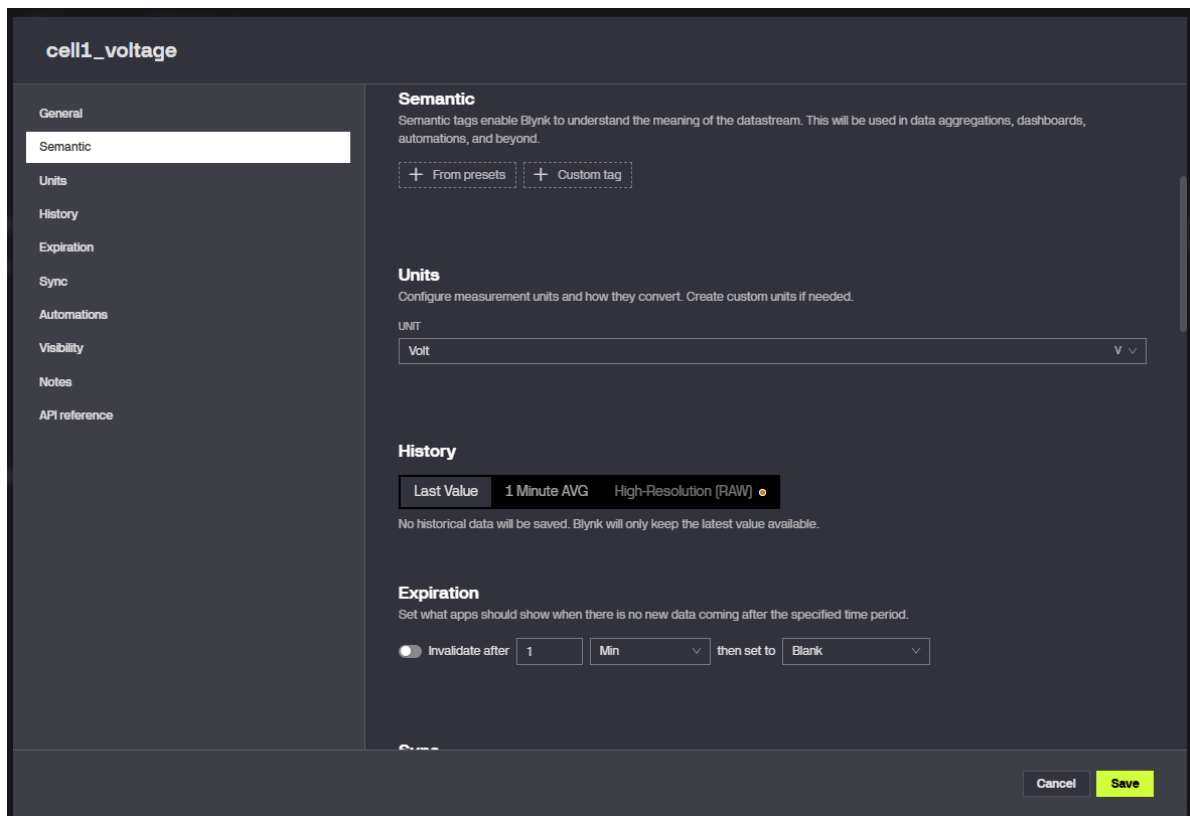


Figure 4. Cell-voltage datastream configuration and available history modes.

The Blynk interface shown during development offered Last Value and 1 Minute AVG history modes, while High-Resolution RAW history was account-dependent. The report therefore does not assume paid raw-history storage.

16. Observed Results

- Live cell-voltage values were displayed on the Blynk dashboard.
- Weakest-cell, strongest-cell and voltage-imbalance values responded to simulated changes.
- Fault conditions produced corresponding fault-state and system-state changes.
- Relay state and relay feedback were monitored independently.
- Composite risk score and battery-health classification changed with system conditions.
- Maintenance recommendation text changed according to the active condition.
- Uptime and fault-event count were visible as dashboard telemetry.
- The custom chart accepted the selected analytics datastreams.
- Dashboard transitions were observed within seconds during the demonstrated tests.

17. Limitations

- The SoC calculation is a simplified voltage-based estimate.
- The prototype uses simulated cell sources and ESP32 ADC behavior; production hardware requires calibrated sensing and protection.
- Blynk historical-data capabilities depend on account/datastream configuration.

- This is a prototype safety/monitoring system, not a certified battery protection system.
- Production deployment would require electrical, thermal, hardware and fault-injection validation.

18. Future Scope

- Chemistry-specific SoC and state-of-health estimation.
- Temperature and current sensing with coulomb counting.
- Dedicated battery-monitoring IC/AFE for accurate cell measurement.
- Authenticated/encrypted production communication.
- Expanded historical analytics and predictive maintenance.
- Hardware-in-the-loop and physical fault-injection testing.

19. Conclusion

The completed prototype demonstrates an end-to-end battery monitoring and safety architecture centered on an ESP32. It combines four-channel voltage monitoring, adaptive imbalance analysis, multi-condition fault detection, a protective state machine, relay feedback verification, offline telemetry buffering and Blynk IoT visualization. The analytics layer converts measured and derived parameters into a composite risk score, battery-health classification and maintenance/operator recommendations. The system therefore provides both local indication through the LCD and remote monitoring through Blynk.

20. Repository and Demonstration

The final submission package should contain the ESP32 source code, documentation, dashboard screenshots and demonstration video. Do not publish the Blynk authentication token in a public repository; use a placeholder/private configuration and regenerate the credential if it has been exposed.

Appendix A — Key Parameters

Parameter	Value
Number of cells	4
ADC valid range	0.0–3.3 V
Over-voltage protection	3.20 V
Under-voltage protection	0.80 V
Over-voltage release	3.10 V
Under-voltage release	0.90 V
Maximum voltage jump	0.50 V
Frozen-sensor tolerance	0.005 V
Offline queue	20 events
Blynk telemetry	1 s
Sensor sampling	1 s
Relay mismatch confirmation	1 s
Recovery verification	5 s

Appendix B — Source Note

This report was prepared from the implemented BMS_PROJECT(1).ino source and the dashboard/configuration screenshots supplied during the project work. It intentionally avoids exposing the Blynk authentication token.

END OF REPORT